

CSA1217- ARTIFICIAL INTELLIGENCE

CO1- ASSESSMENT TOOL 2

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Question 1: Doctor Shift Scheduling Using Backtracking

Problem Statement:

Assign three doctors (D1, D2, D3) to Morning, Afternoon and Night shifts while satisfying all given constraints.

Constraints

- D1 cannot work the Night shift.
- D2 must work before D3 (Morning < Afternoon < Night).
- Only one doctor can be assigned to each shift.
- D3 cannot work the Morning shift.
- Each doctor must be assigned exactly one shift.

Backtracking Steps

1. Assign D1 = Morning ✓
2. Assign D2 = Afternoon ✓ (before D3)
3. Assign D3 = Night ✓
4. Check Constraint: D3 is not in Morning ✓
5. Check Constraint: D1 is not in Night ✓
6. Check Constraint: One doctor per shift ✓
7. All constraints satisfied. Solution accepted.

Final Valid Schedule

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Question 2: Robot Grid Navigation using BFS

The robot moves in a 5×5 grid from Start (S) to Goal (G). It can move Up, Down, Left and Right. Obstacles cannot be crossed and each move costs 1.

Algorithm Used

Breadth-First Search (BFS)

Step-by-Step Expansion

Step 1: Insert Start node S into the queue.

Step 2: Remove S and expand its valid neighbouring cells.

Step 3: Add all unvisited neighbours into the queue.

Step 4: Continue expanding nodes level by level.

Step 5: Ignore blocked cells and already visited nodes.

Step 6: Stop when Goal node G is reached.

Priority Queue / Queue Updates

Queue: [S] → [A,B] → [B,C,D] → [C,D,E] → ... → [G]

Optimal Path

S → A → C → E → G (Example)

Total Cost

Cost = Number of moves × 1

Question 3: Autonomous Rescue Robot

a) Constraint Analysis

- The robot operates in a partially observable environment because it can only observe nearby cells.
- Obstacles prevent direct movement and require alternative paths.
- Risky zones increase the path cost by 2, so the robot avoids them whenever possible.
- The robot continuously updates its knowledge while exploring.
- The objective is to safely reach the survivor with minimum total cost.

b) Solution Approach

Uniform Cost Search (UCS) combined with an Online Search Agent is the most suitable approach.

8. Start from S.
9. Observe neighbouring cells.
10. Expand the lowest-cost path first.
11. Update the map whenever new information is discovered.
12. Avoid obstacles and high-cost risky zones whenever possible.
13. Reach Goal (G) with minimum cumulative cost.

c) AI Concepts Applied

- Intelligent Agent: The robot senses, plans and acts autonomously.
- Rationality: It always chooses the safest and least-cost path.
- Environment Type: Partially observable, dynamic, sequential and uncertain.

d) Justification

Uniform Cost Search guarantees the least-cost path, while the Online Search Agent allows the robot to adapt to newly discovered obstacles and risks. This combination is ideal for disaster environments where complete information is unavailable.